Physics in VJEPA2

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Data splits

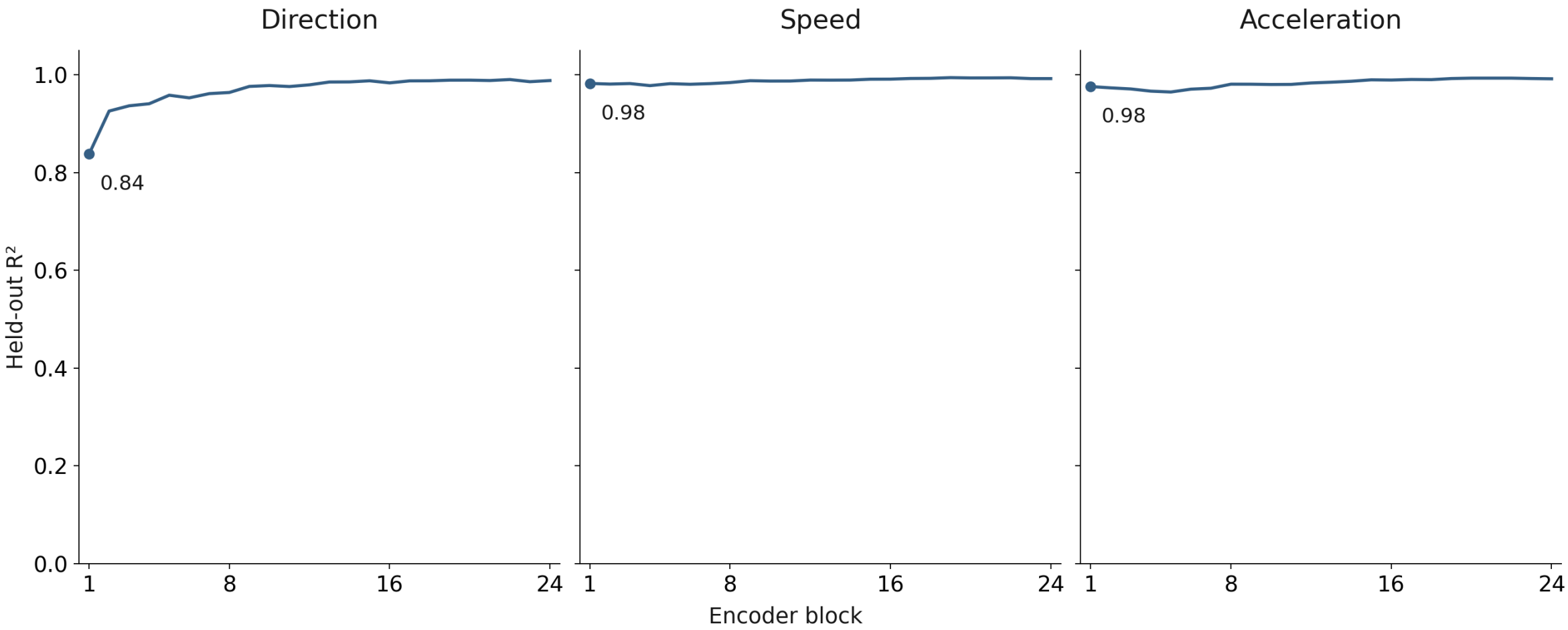
Disjoint video splits within each physical variable

Split	Purpose	Direction	Speed / accel.
A	Fit probes and splines	768	768
B	Select hyperparameters	220	256
C	Train independent readouts	220	256
D	Evaluate probes and edits	292	256

Clips per variable. Follow-up analyses reuse D and are exploratory.

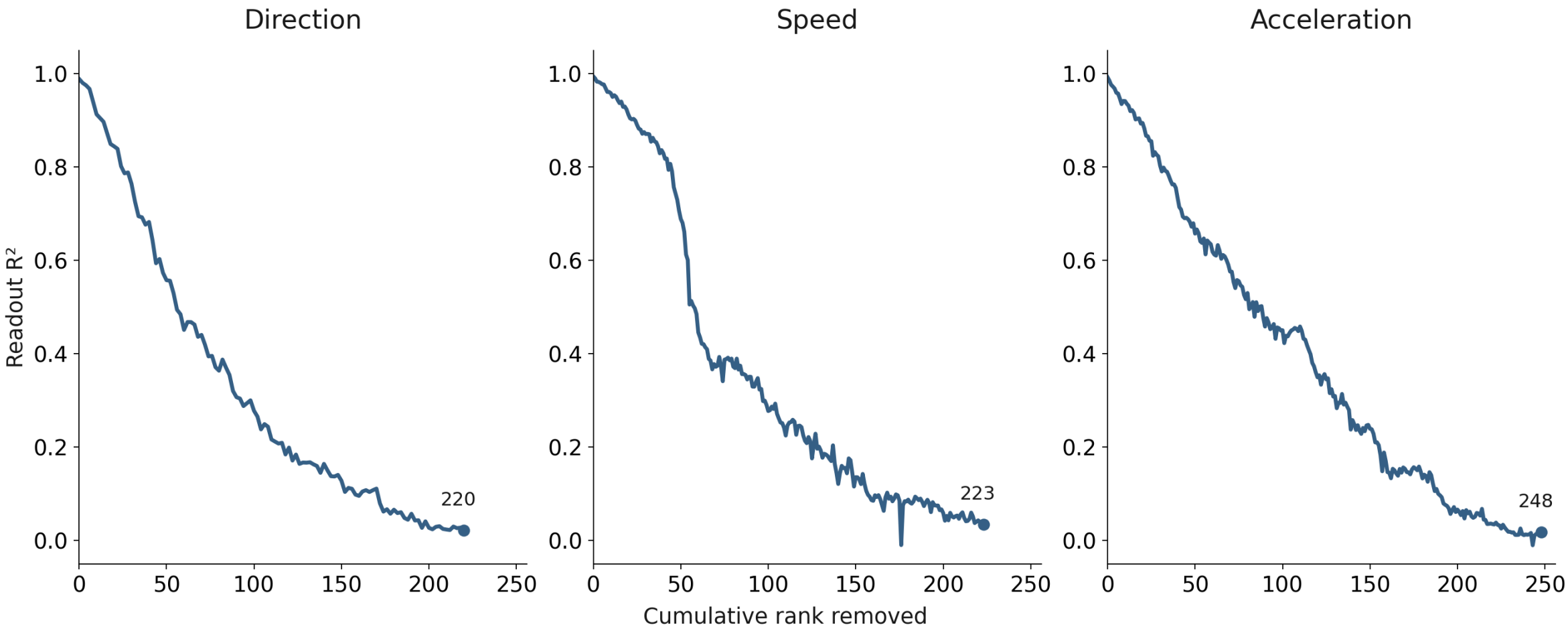
Linear probing across layers

No distinct physics emergence zone on these clips: all three variables are readable from block 1.



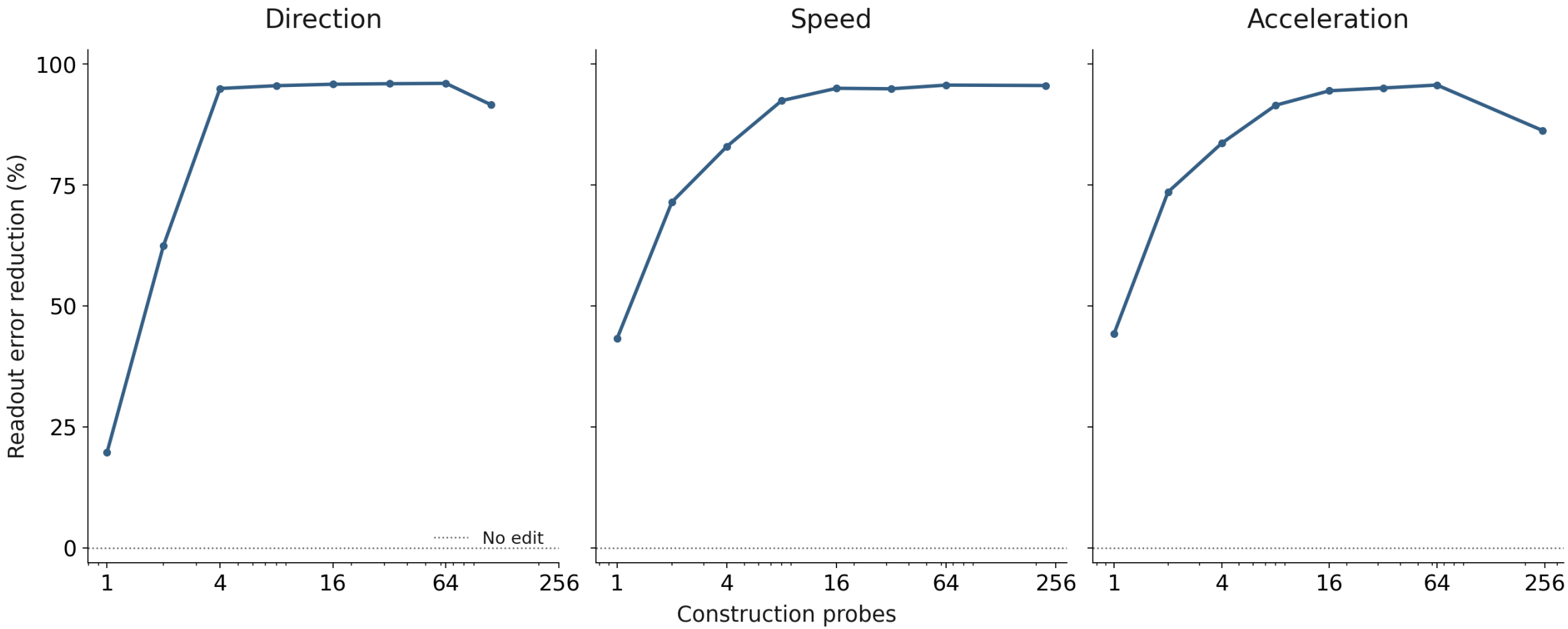
Iterative nullspace probes

Fresh linear readout after each removal



Multi-probe steering

Readout error reduction vs no edit · higher is better
0% = no improvement · 100% = target reached



Readout agreement does not establish causality

Edits move an independently trained probe’s estimates toward the target.

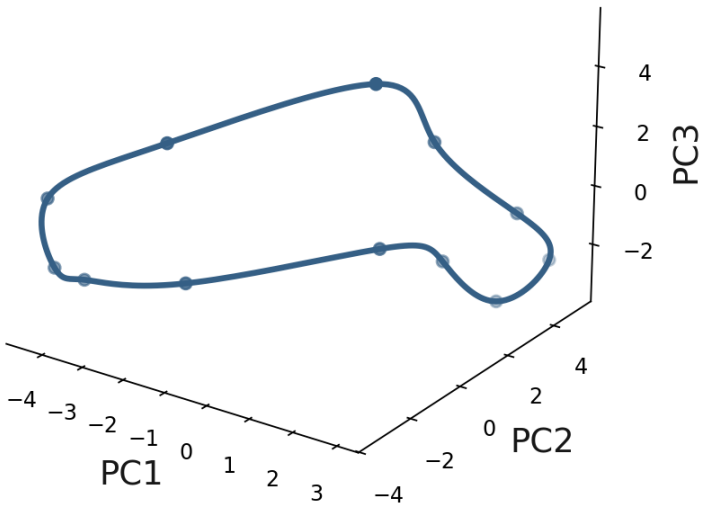
Both probes can use the same correlated features. Their agreement does not show that the predictor uses those features.

Causal test: edit velocity, run the predictor, and measure the change in predicted position.

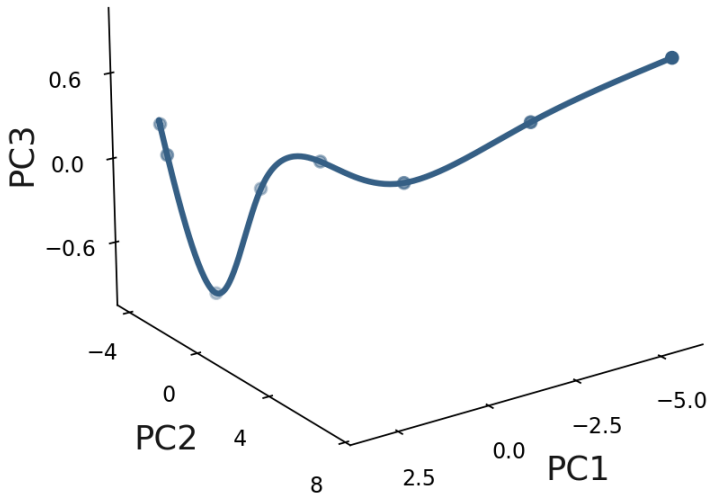
Centroid splines

64-dimensional PCA fit; first three components shown

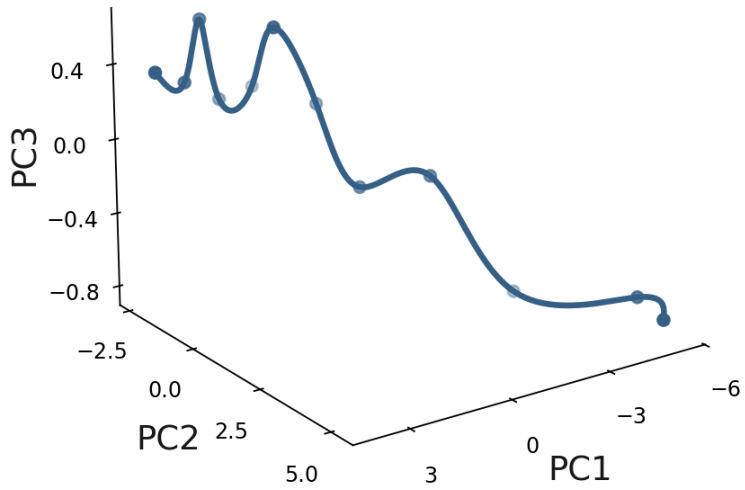
Direction (periodic)



Speed



Acceleration



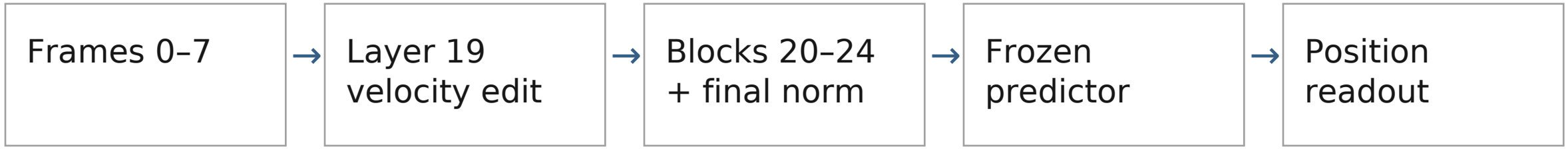
Spline versus linear steering

Independent-readout error reduction vs no edit · higher is better
0% = no improvement · 100% = target reached

Method	Direction	Speed	Acceleration
Spline	94.3%	87.8%	89.6%
Linear (16 probes)	95.9%	95.0%	94.5%

Steering through the predictor

Earlier: one full-clip pooled vector → external probe
Here: 1,024 prefix tokens → remaining encoder + predictor



Fit a probe per token and make token-specific edits.

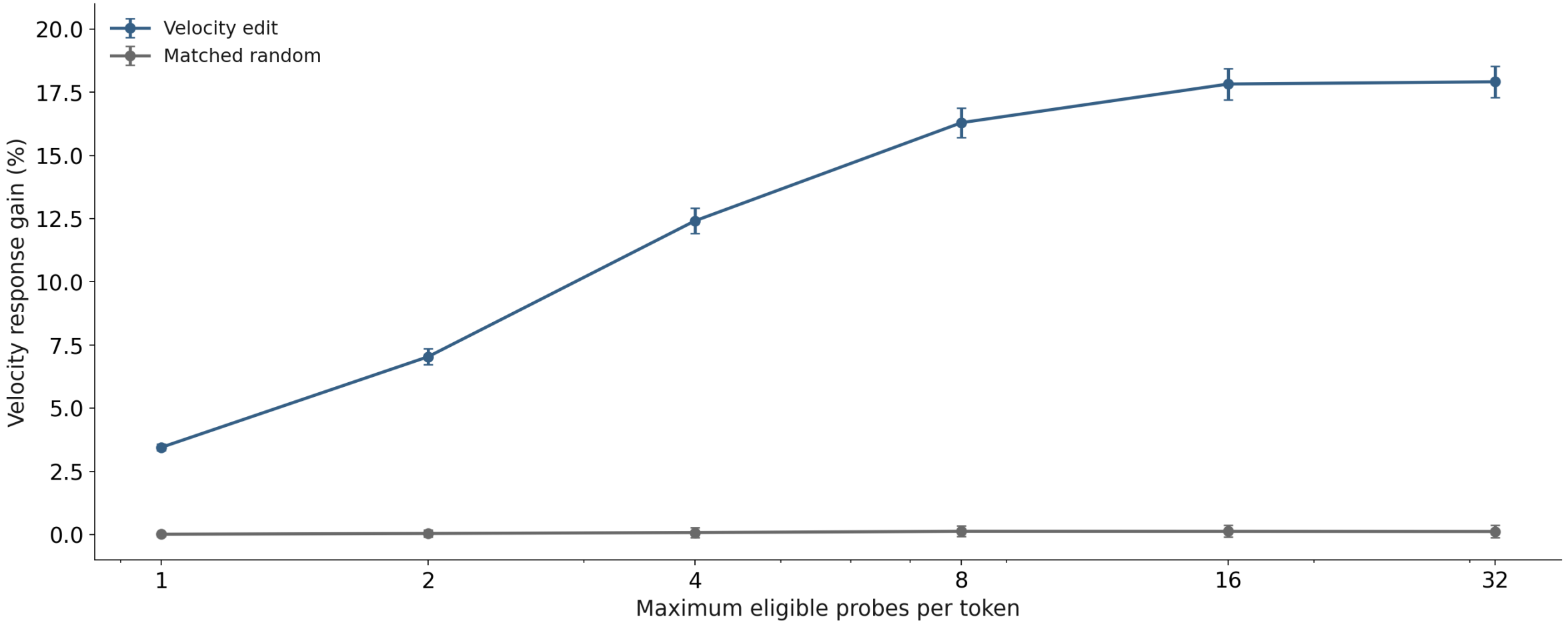
Future targets: frames 8–15

Expected displacement: $\Delta p(t) = \Delta v \cdot (t - t_0)$

Response gain = measured velocity change / requested change

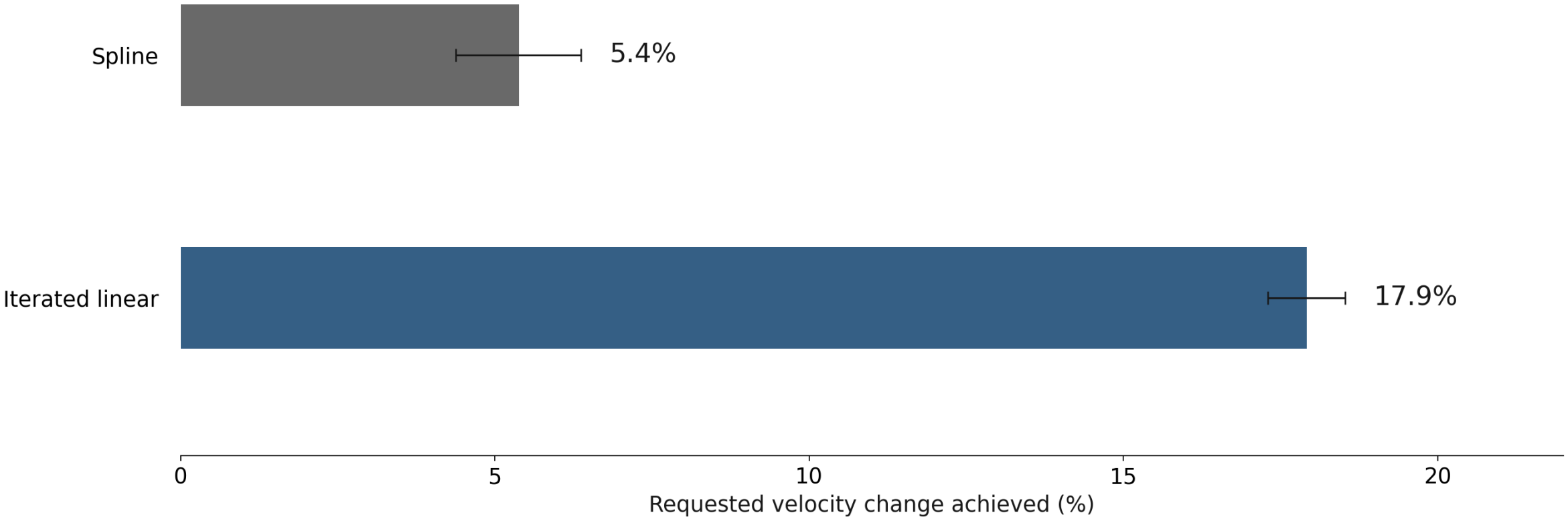
Iterated velocity steering

Δspeed = +0.50 m/s · 184 D clips · 95% bootstrap intervals



Predicted motion: spline versus linear

Requested +0.50 m/s · 184 D clips · 95% bootstrap intervals



Conclusions

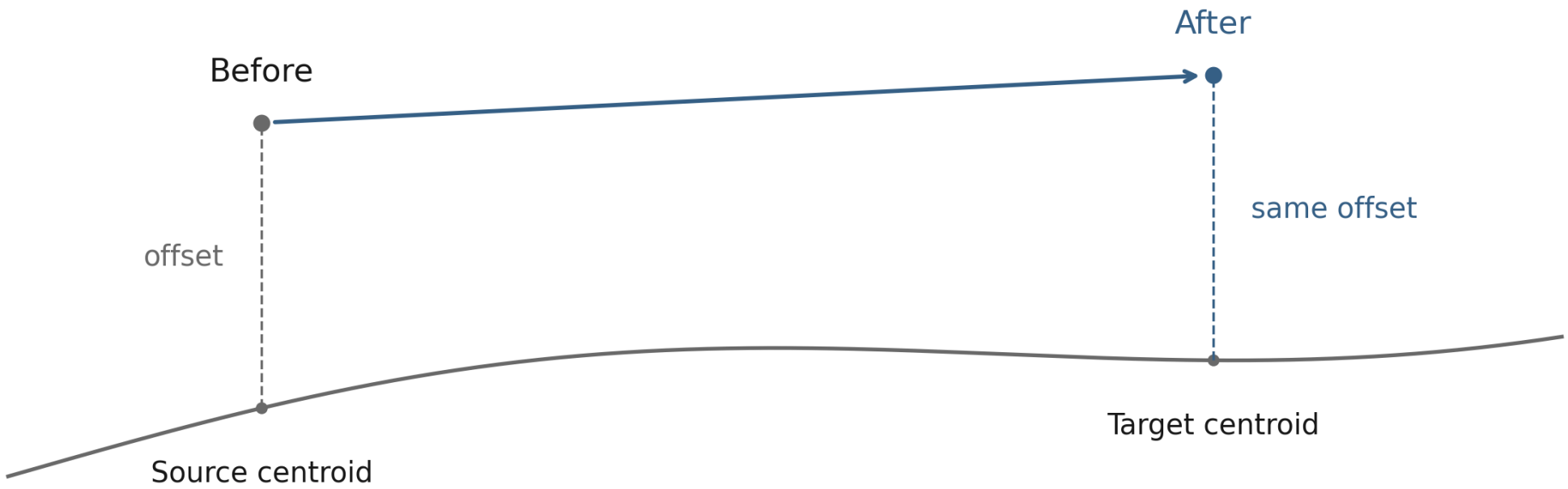
1. All three variables are accurately and redundantly represented.

2. Independent-readout accuracy alone does not establish causal control.

3. Iterated linear edits produce the strongest measured motion response.

Spline steering preserves the offset

Add the source-to-target centroid difference · known source label · schematic PCA space



$$z' = z + f(\text{target}) - f(\text{source})$$

Position-readout limitations

Unedited baseline / diagnostic	Farthest D error	What it tells us
Predictor → C-trained position ridge	8.17 px	Beats holding the last position
Hold last observed position	21.26 px	Static reference
Measured-prefix extrapolation	0.78 px	Simple physics remains much better

Baselines: 256 D clips.

Spline curvature comparison

Offset-preserving edits · interior waypoints · readout error reduction vs no edit

